biohub-cell-tracking-during-development

6th Place Solution: A Tracker That Teaches Itself

Rank	6
Team	Cyrus
Author	Cyrus
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Source: https://www.kaggle.com/competitions/biohub-cell-tracking-during-development/discussion/745676
Retrieved with Kaggle CLI on 2026-10-04.

Thanks to Biohub and Kaggle for a great competition and a hard, interesting dataset.

Write-up: https://www.kaggle.com/competitions/biohub-cell-tracking-during-development/writeups/6th-place-solution Code: https://github.com/jamal-saeedi/biohub_cell_tracking_kaggle

Our final submission scored 0.953 private (0.961 public). A second submission with the same pipeline and best-validation checkpoints scored 0.955 private.

The approach in brief

One network detects and links. Each model reads three frames at native resolution. It predicts cell centres, and for every cell a probability over its candidate parents in the previous frame, plus an explicit "new cell" class. The linker uses features from the same image features that found the cells, so it can tell similar-looking neighbours apart. Six models, one set of cells. Three IsotropicLineageNet and three MultiScaleLineageNet models are averaged into one set of cells. Each model then scores the same candidate links with its own head and its own LightGBM re-scorer, and the six link probabilities are averaged. A global ILP turns cells and links into tracks and divisions. The link cost includes a distance corrected for microscope stage drift. A drift-compensated smoother then refines the positions. Training on 3 % annotation: Detection targets mark three states: positive, verified background, and unknown, which carries no loss. A count prior pulls the predicted number of cells towards the organisers' estimate. In each teacher → student round, the teacher is the whole pipeline. Its tracks on the training movies become pseudo-labels for the next models; there were 7 label sets in all. What mattered most:

test-time augmentation, 1 → 4 views on a single model: +0.05; the drift-corrected link distance: +0.009 on validation; the first teacher → student round: +0.008 on validation; member-own re-scorers: +0.004 to +0.007 per model; ensembles of models on different splits and architectures: 0.954 → 0.959 public; fp16 instead of emulated bf16 on the T4: 3.5× faster prediction, which made the six-model ensemble fit in the 12 hours. Everything is public:

Models (MIT): https://www.kaggle.com/models/jamalsaeedi/biohub-cell-tracking Inference notebook: https://www.kaggle.com/code/jamalsaeedi/biohub-cell-tracking-inference. It reproduces both submissions byte for byte. Code dataset: https://www.kaggle.com/datasets/jamalsaeedi/biohub-cell-tracking-kaggle The GitHub repo also has: the full training code, with one recipe per model of the lineage; pseudo-label generation and re-scorer training; a CPU smoke test and a training notebook. Happy to answer questions, and congratulations to all the winners!